

Question ID: f678483b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

A submersible device is used for ocean research. The function $g(x) = -\frac{1}{55}(x + 19)(x - 35)$ gives the depth below the surface of the ocean, in meters, of the submersible device x minutes after collecting a sample, where $x > 0$. How many minutes after collecting the sample did it take for the submersible device to reach the surface of the ocean?

Correct Answer: 35

Rationale

The correct answer is **35**. It's given that the function $g(x) = -\frac{1}{55}(x + 19)(x - 35)$ gives the depth below the surface of the ocean, in meters, of the submersible device x minutes after collecting a sample, where $x > 0$. It follows that when the submersible device is at the surface of the ocean, the value of $g(x)$ is **0**. Substituting **0** for $g(x)$ in the equation $g(x) = -\frac{1}{55}(x + 19)(x - 35)$ yields $0 = -\frac{1}{55}(x + 19)(x - 35)$. Multiplying both sides of this equation by -55 yields $0 = (x + 19)(x - 35)$. Since a product of two factors is equal to **0** if and only if at least one of the factors is **0**, either $x + 19 = 0$ or $x - 35 = 0$. Subtracting **19** from both sides of the equation $x + 19 = 0$ yields $x = -19$. Adding **35** to both sides of the equation $x - 35 = 0$ yields $x = 35$. Since $x > 0$, **35** minutes after collecting the sample the submersible device reached the surface of the ocean.